



Bangladesh University of Business and Technology

Department of Computer Science and Engineering

PROPOSAL OF CampusConnect

A Smart Student Community & Academic Support Platform

Course Title: Software Development IV

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1. Introduction

University students face numerous academic and non-academic challenges throughout their academic journey. Study materials, CT question banks, and senior notes are scattered across different platforms and hard to find when needed. Tasks like creating lab report covers or assignment covers are done manually and waste valuable time. Job and internship opportunities are not easily reachable, and emergency situations like blood donation requests often fail to reach the right people in time. Club events go unnoticed, lost items go unreported, and juniors struggle to find affordable books from seniors. On top of that, fake accounts in online groups reduce trust among students.

To solve these problems, this project proposes **CampusConnect** — a web-based platform built exclusively for university students. It brings all essential student services together in one place, including a CT question bank and senior notes repository, academic document generators, a job board, emergency blood donation support, club and event management, a lost and found system, emergency contacts, and a book marketplace for seniors to sell books to juniors. The platform uses student ID and university email verification to ensure a safe and trusted community.

CampusConnect aims to reduce dependency on multiple fragmented platforms and improve student productivity, communication, and overall campus experience.

2. Related Works

Students currently rely on university portals, social media groups, messaging applications, and external websites for academic and community-related services. While these platforms offer some useful functionalities, they lack integration and often fail to provide a complete student-oriented solution.

Most university management systems focus exclusively on academic records and administrative tasks. Community-based features such as resource sharing, emergency support, book exchange, and campus life management remain unavailable. CampusConnect aims to bridge this gap by integrating multiple student services into a unified and dedicated platform.

3. Objectives

The primary objective of CampusConnect is to develop a centralized platform that improves student productivity, communication, and collaboration. The specific objectives are:

- Maintain a centralized repository of study materials, notes, and CT question banks (*CT Question Bank + Senior Notes & Suggestion*)
- Enable students to generate academic documents instantly (*Lab Report Cover + Assignment Cover*)
- Connect students with job and internship opportunities (*Job & Internship Board*)
- Support emergency blood donation requests within the campus community (*Emergency Blood Donation*)
- Improve campus communication through clubs and event management (*Club & Event Hub*)
- Help students report and find lost belongings (*Lost & Found*)

- Provide all emergency contact numbers in one place (*Emergency Contact*)
- Enable juniors to buy previous semester books from seniors (*Book Marketplace*)
- Provide a secure platform through student identity verification
- Enhance overall campus engagement and student support

4. Problem Statement

University students commonly face the following challenges that necessitate a centralized solution:

- Study materials, notes, and CT questions are scattered and hard to find
- Creating lab report and assignment covers manually is time-consuming
- Job and internship opportunities are not easily accessible for students
- Emergency blood donation requests often fail to reach potential donors on time
- Club notices and event information have no dedicated centralized place
- Lost belongings have no structured reporting or tracking system
- Emergency contact numbers are not available in one place
- Juniors have no platform to buy affordable used books from seniors
- Fake accounts reduce trust within online student communities

5. Methodology

5.1 Technology Stack

Layer	Technologies
Frontend	React.js, HTML5, CSS3, Tailwind CSS
Backend	Django, Django REST Framework
Database	PostgreSQL
Authentication	Student ID Verification, University Email Verification, JWT

5.2 Development Process

The project will follow an iterative development process consisting of the following phases:

1. Requirement Analysis
2. System Design
3. Database Design
4. Frontend Development
5. Backend Development
6. Authentication Integration
7. Testing and Debugging
8. Deployment

6. System Features

CampusConnect is organized into four core feature categories, each designed to address a specific dimension of student life:

Academic Features	Career Features
Lab Report Cover Generator	Job & Internship Board
Assignment Cover Generator	
CT Question Bank	
Senior Notes & Suggestion	
Community Features	Security & Marketplace
Lost & Found	Book Marketplace (Senior to Junior)
Emergency Contacts	Student ID Verification
Club & Event Hub	University Email Verification
Emergency Blood Donation	

7. Expected Outcomes

The successful implementation of CampusConnect will deliver the following measurable outcomes:

- ◇ A centralized academic resource platform with easy access to notes and CT question banks
- ◇ Instant generation of academic documents including lab report and assignment covers
- ◇ A structured job and internship board accessible to all students
- ◇ Improved emergency support services including blood donation coordination
- ◇ Increased student participation in campus clubs and events
- ◇ A reliable lost and found system for campus belongings
- ◇ A trusted book marketplace allowing juniors to buy affordable books from seniors
- ◇ A secure, verified student community free from fake accounts

8. Conclusion

CampusConnect is designed to become a comprehensive student ecosystem that combines academic support, document generation tools, career opportunities, community services, and a book exchange marketplace into a single platform. By reducing dependence on multiple fragmented applications and ensuring a secure environment through robust student verification, the system will significantly improve student engagement, resource sharing, and overall campus connectivity.

The project has strong practical value and the potential to be implemented across universities as a dedicated student support platform. It addresses real, pressing challenges faced by students daily and offers a scalable, modern solution built on industry-standard technologies.

9. Future Scope

Future enhancements planned for the CampusConnect platform include:

- AI Academic Assistant for personalized learning support
- Mobile Application for Android and iOS platforms
- CGPA Calculator and academic progress tracker
- Smart Recommendation System for resources and opportunities
- Academic Analytics Dashboard
- Real-Time Notification System
- Inter-University Collaboration Network
- Bus Tracker with real-time location
- Campus Map with indoor navigation

10. References

1. Django Documentation — <https://docs.djangoproject.com>
2. React Documentation — <https://react.dev>
3. Django REST Framework Documentation — <https://www.django-rest-framework.org>
4. PostgreSQL Documentation — <https://www.postgresql.org/docs>
5. JWT Authentication Documentation — <https://jwt.io/introduction>
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